While that task is far from complete, one implication seems clear: the simple picture sketched in the last chapter, where space emerges from a set of entangled nearest-neighbor degrees of freedom, is probably not the entire story. It's a very good story, and might be the right starting point for constructing a theory of quantum gravity. But it relies heavily on the idea of locality—what happens at one point in space can have an immediate effect only on points right next door. Black holes, to the extent that we understand them, seem to be indicating that nature is more subtle than that. In some circumstances the world looks like a collection of degrees of freedom interacting with their nearest neighbors, but when gravity becomes strong, that simple picture breaks down. Rather than being distributed throughout space, degrees of freedom squeeze together on a surface, and "space" is merely a holographic projection of the information contained therein.

Locality undoubtedly plays an important role in our everyday lives, but it seems like the fundamental nature of reality can't quite be captured by a set of things happening at precise locations in space. Once again, what we have here is a job for the Many-Worlds approach to quantum mechanics. Other formulations take space as a given and work within it; the wave-function-first Everettian philosophy allows us to accept that space can appear fundamentally different depending on how we look at it, if it's a useful concept at all. Physicists are still wrestling with the implications of this idea, but it's already led us to some very interesting places indeed.

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In general relativity, a black hole is a region of spacetime that is curved so dramatically that nothing can escape from it, not even light itself. The edge of the black hole, demarcating the inside from the outside, is the *event horizon*. According to classical relativity, the area of the event horizon can only grow, not shrink; black holes increase in size when matter and energy fall in, but cannot lose mass to the outside world.

Everyone thought that was true in nature until 1974, when Hawking announced that quantum mechanics changes everything. In the presence of quantum fields, black holes naturally radiate particles into their surroundings. Those particles have a blackbody spectrum, so every black hole has a temperature; more massive black holes are cooler, while very